Educational Futures of Intelligent Synergies Between Humans, Digital Twins, Avatars, and Robots - the ISTAR Framework

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Abstract

With the rapid advances of Artificial Intelligence (AI) and its technologies, human teachers and machines are now capable of collaborating to effectively achieve specified outcomes. In educational settings, such collaboration requires consideration of several dimensions to ensure safe, responsible, and ethical usage. While various research studies have discussed human-machine collaboration or cooperation in education, a framework is now needed that aligns with contemporary affordances. Providing such a framework can help to better understand how human teachers and machines can team up in education and what should be considered while doing so. To address this gap, this paper outlines the iSTAR (Intelligent human-machine Synergy in collaborative teaching: utilizing the digital Twins, Avatars/Agents and Robots) framework. iSTAR represents human-machine collaboration as an ecosystem that goes beyond the simple collaboration between human teachers and machines in education. Therefore, it presents core dimensions of DELTA (design, ethics, learning, teach-

ing and assessments) that should be considered in designing safe, responsible, and ethical learning opportunities.